MATH 2410 · PROBABILITY THEORY · AUTUMN 2026

Problem Set 4

Conditioning, Densities and Expectation

Due **Friday 30 October, 17:00** · Dr. Imogen Hale

PROBLEM	POINTS	SCORE
1	10	
2	12	
3	10	
4	8	
5	10	
TOTAL	50	

STUDENT NAME

STUDENT ID

Show your reasoning: a correct number without justification earns at most half the points. Collaboration is welcome, but write up your own solutions and name the people you worked with.

10 pts

PROBLEM 1 | *A screening test*

A blood test for a condition that affects 2% of a population detects it in 95% of people who have it (sensitivity) and correctly clears 90% of people who do not (specificity). A person is drawn at random and tested.

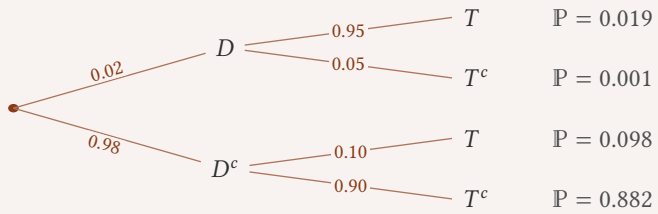
(a) Draw the probability tree for condition and test result.

(b) Find the probability that the test is positive.

(c) Given a positive result, how likely is it that the person has the condition? Comment on the size of your answer.

SOLUTION

(a) Let D be the event that the person has the condition and T that the test is positive. The tree branches first on D, then on T:



(b) By the law of total probability, summing the two leaves that end in T,

$$\begin{aligned}\mathbb{P}(T) &= \mathbb{P}(T \mid D)\,\mathbb{P}(D) + \mathbb{P}(T \mid D^c)\,\mathbb{P}(D^c) \\ &= 0.95 \cdot 0.02 + 0.10 \cdot 0.98 = 0.019 + 0.098 = 0.117.\end{aligned}$$

(c) Bayes' rule gives

$$\mathbb{P}(D \mid T) = \frac{\mathbb{P}(T \mid D)\,\mathbb{P}(D)}{\mathbb{P}(T)} = \frac{0.019}{0.117} \approx 0.162.$$

Only about one positive in six is a true case. The condition is rare, so the 10% false-positive rate among the healthy 98% produces five times more positives than the sick 2% do.

12 pts

PROBLEM 2 | *A polynomial density*

A random variable X has density $f(x) = c\,x^2$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere.

(a) Find the constant c.

(b) Find the distribution function $F(x)$ and use it to compute $\mathbb{P}(X > 1)$.

(c) Compute $\mathbb{E}[X]$ and $\operatorname{Var}(X)$.

SOLUTION

(a) The density must integrate to one: $\int_0^2 c\,x^2\,dx = \frac{8c}{3} = 1$, so $c = \frac{3}{8}$.

(b) For $0 \le x \le 2$, $F(x) = \int_0^x \frac{3}{8}t^2\,dt = x^3/8$, with $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 2$. Hence $\mathbb{P}(X > 1) = 1 - F(1) = \frac{7}{8}$.

(c) The first two moments are

$$\mathbb{E}[X] = \int_0^2 x \cdot \frac{3}{8}x^2\,dx = \frac{3}{8}\cdot\frac{2^4}{4} = \frac{3}{2},$$
$$\mathbb{E}[X^2] = \int_0^2 x^2 \cdot \frac{3}{8}x^2\,dx = \frac{3}{8}\cdot\frac{2^5}{5} = \frac{12}{5},$$

so that

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{12}{5} - \frac{9}{4} = \frac{3}{20}.$$

The mean lies right of the midpoint 1, as expected for a density that grows with x.

10 pts

PROBLEM 3 | *Waiting for a success*

A basketball player makes each free throw independently with probability $p \in (0,1)$. Let N be the number of throws up to and including the first success.

(a) Show that $\mathbb{P}(N > n) = (1-p)^n$ for $n = 0, 1, 2, \dots$

(b) Prove the memoryless property $\mathbb{P}(N > m + n \mid N > m) = \mathbb{P}(N > n)$.

(c) Use $\mathbb{E}[N] = \sum_{n \ge 0} \mathbb{P}(N > n)$ to find $\mathbb{E}[N]$.

SOLUTION

(a) The event $\{N > n\}$ happens exactly when the first n throws all miss. By independence its probability is $(1-p)^n$.

(b) Because $\{N > m+n\} \subseteq \{N > m\}$, the intersection of the two events is the smaller one, and

$$\mathbb{P}(N > m+n \mid N > m) = \frac{\mathbb{P}(N > m+n)}{\mathbb{P}(N > m)} = \frac{(1-p)^{m+n}}{(1-p)^m}$$
$$= (1-p)^n = \mathbb{P}(N > n),$$

so past misses say nothing about the wait still to come.

(c) Summing the geometric series,

$$\mathbb{E}[N] = \sum_{n \ge 0} (1-p)^n = \frac{1}{1-(1-p)} = \frac{1}{p}.$$

A player who scores 80% of the time waits 1.25 throws on average.

8 pts

PROBLEM 4 | *Returned coats*

At the end of a concert, an attendant hands n coats back to their n owners in uniformly random order. Let M be the number of people who receive their own coat.

(a) Write M as a sum of indicator variables and find $\mathbb{E}[M]$.

(b) Find $\operatorname{Var}(M)$.

SOLUTION

(a) Let $I_k = 1$ if person k gets their own coat and 0 otherwise, so $M = I_1 + \cdots + I_n$. Each person is equally likely to get any of the n coats, so $\mathbb{E}[I_k] = 1/n$, and linearity of expectation gives $\mathbb{E}[M] = n \cdot \frac{1}{n} = 1$, whatever the size of the audience.

(b) For $j \neq k$, $\mathbb{E}[I_j I_k] = \frac{1}{n} \cdot \frac{1}{n-1}$, the chance that both get their coats. Since $I_k^2 = I_k$,

$$\mathbb{E}[M^2] = \sum_k \mathbb{E}[I_k] + \sum_{j \neq k} \mathbb{E}[I_j I_k] = 1 + n(n-1) \cdot \frac{1}{n(n-1)} = 2,$$

$$\operatorname{Var}(M) = \mathbb{E}[M^2] - \mathbb{E}[M]^2 = 2 - 1 = 1.$$

Mean and variance both equal one, a first hint that M is close to a Poisson(1) variable when n is large.

10 pts

PROBLEM 5 | *Two dice, two summaries*

Roll two fair dice. Let X be the smaller and Y the larger of the two faces, so that $X \le Y$ always.

(a) Find $\mathbb{P}(X = j,\ Y = k)$ for $j \le k$, and then $\mathbb{P}(X = k)$.

(b) Find $\mathbb{E}[X]$, and deduce $\mathbb{E}[Y]$ without further sums.

(c) Are X and Y independent? Justify in one line.

SOLUTION

(a) The outcome $(X, Y) = (j, k)$ with $j < k$ arises from two ordered rolls, (j, k) and (k, j), while $j = k$ arises from one, so $\mathbb{P}(X = j, Y = k)$ is $2/36$ or $1/36$. Summing over the $6 - k$ larger values of Y,

$$\mathbb{P}(X = k) = \frac{1 + 2(6 - k)}{36} = \frac{13 - 2k}{36}, \qquad k = 1, \dots, 6.$$

(b) Using $\sum_{k=1}^{6} k = 21$ and $\sum_{k=1}^{6} k^2 = 91$,

$$\mathbb{E}[X] = \sum_{k=1}^{6} k \, \frac{13 - 2k}{36} = \frac{13 \cdot 21 - 2 \cdot 91}{36} = \frac{91}{36} \approx 2.53.$$

Since $X + Y$ is the sum of the two faces, $\mathbb{E}[X] + \mathbb{E}[Y] = 7$, so $\mathbb{E}[Y] = 7 - \frac{91}{36} = \frac{161}{36} \approx 4.47$.

(c) No: $\mathbb{P}(X = 2,\ Y = 1) = 0$, but $\mathbb{P}(X = 2)\,\mathbb{P}(Y = 1) > 0$.